

WORKING DRAFT

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Todo list

☐ Citation needed.	5
☐ Add appendix showing questions selected.	17



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**Aligning Large Language Models to
publicly collected values**

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Submitted in partial fulfilment of the requirements for
Master of Artificial Intelligence.

Abstract

A short description of the project goes here.

Acknowledgments

Any acknowledgments should go in here, between the title page and the table of contents. The acknowledgments do not form a proper chapter, and so don't get a number or appear in the table of contents.

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Chapter 1

Introduction

This chapter gives an introduction to the project report.

5-6 pages with a full summary

1.1 Report overview

Following on from this introduction chapter there are 5 more chapters to the report. The next chapter 2 gives the key background

This report is structured into 3 main chapters, each focusing on a specific part of the project. For each chapter of the project it explains the methods used and the results obtained. Each chapter is self contained and can be read somewhat independently of the others. The 3 main chapters are:

- 3: Fine-tuning of the LLM on WVS data
- 4: Behavioural analysis of the fine-tuned and base LLMs
- 5: Public consultation on the alignment of the LLMs

The final chapter 6 concludes the report with a summary of the findings, conclusions drawn from the project, key limitations and recommendations for future work.

Chapter 2

Background

A related works section.

Could be removed from word count.

Includes a biref (few pages) summary of the important bits of the related works and how they relate to this project.

This work can understand from multiple perspectives. Here are some
survey modelling [2] [6]
annotation disagreement [4]
student teacher and distillation

Chapter 3

Fine-tuning

This chapter will talk about the fine-tuning process used.

clustering of WVS data prompts and format of the data finetuning methods finetuning evaluations (survey modelling and response distributions)

The goal of this chapter is to show that an open weight model can be fine-tuned with value survey data to produce a model that responds more like the respondents it is aligned to when answering survey questions.

In section 3.1 we will find that ...

In section 3.2 and section 3.3 I describe the methods used to create the dataset and fine-tune the model.

3.1 The results of Fine-tuning LLMs on value survey data

3.1.1 The clusters of survey respondents

The WVS data was clustered into 2 clusters using Latent Cluster Analysis (LCA). The clusters were found to be stable and reproducible. We can understand the cluster in two ways. Firstly we can look at the value questions that most split the clusters. Secondly we can look at the demographic questions that most split the clusters.

For the value questions in Figure 3.1 we can see that the top 5 value questions are interestingly around trust in authority (elections and World Health organisation), homosexuality and immigration.

For the demographic questions in Figure 3.2 we can see that one group (cluster_1) is an older, more religious, less educated and poorer group. This fits in with some standard social science observation .

As is seen in Figure 5.1 it is possible to assign respondents to cluster with a high degree of certainty (>90%) with only 10 questions. This is important for the methodology of the public consultation survey.

Citation needed.

3.1.2 Efficacy of fine-tuning

Discuss the various methods of evaluating the efficacy of fine-tuning, including: - Survey modelling (how well the model predicts survey responses) - Response distributions (how well the model's response distributions match the survey data)

Also explore other concepts of: - Is fine-tuning robust to reasoning on and off? - Does system prompt matter?

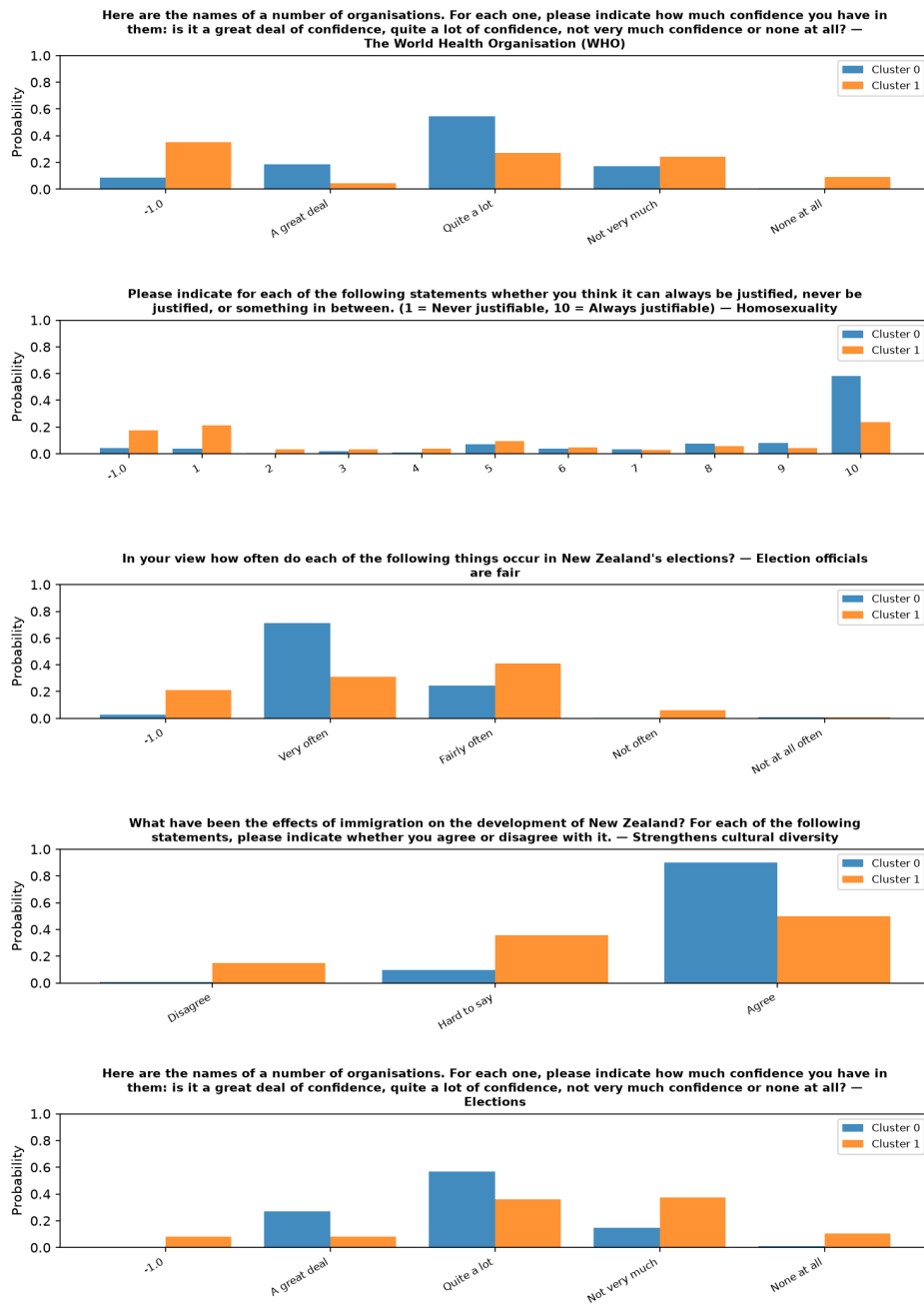


Figure 3.1: The top 5 value questions that provide the most amount of information about which cluster a respondent belongs to.

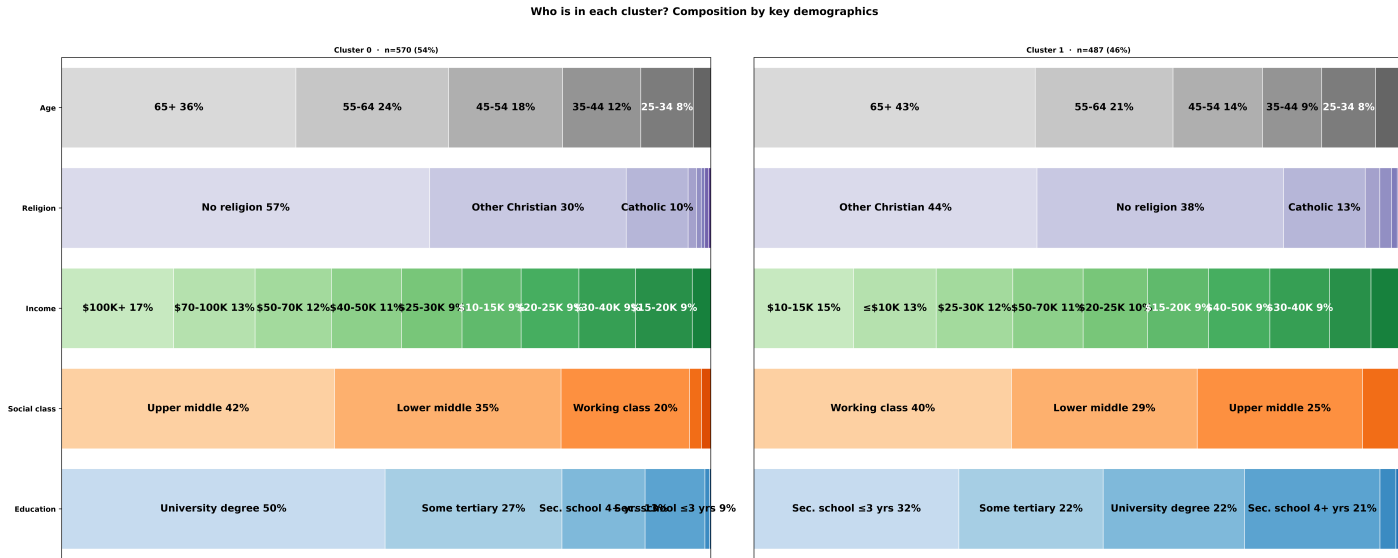


Figure 3.2: The demographic differences between the clusters.

Model	Accuracy			Cross-entropy			KL divergence		
	train	validation	overall	train	validation	overall	train	validation	overall
Base	47.8	58.2	48.2	2.02	1.99	2.01	0.880	0.879	0.879
Modal response (cluster 0)	47.6	58.0	47.8	1.92	1.84	1.92	0.877	0.851	0.877
Modal response (cluster 1)	41.5	41.0	42.2	2.15	2.18	2.14	0.961	1.001	0.958
Modal response (overall)	47.9	58.4	48.5	2.02	1.99	2.01	0.879	0.881	0.879
Sampled response (cluster 0)	46.8	58.2	47.4	1.91	1.84	1.91	0.874	0.855	0.874
Sampled response (cluster 1)	41.5	41.1	42.3	2.14	2.18	2.14	0.958	1.002	0.958
Sampled response (overall)	47.7	58.4	48.3	2.02	2.00	2.02	0.882	0.886	0.879
First token (overall)	47.6	58.3	48.2	2.02	1.99	2.01	0.879	0.884	0.879
Full string (overall)	47.7	58.4	48.3	2.02	1.99	2.01	0.880	0.881	0.879

Table 3.1: The results of fine-tuning Qwen-3.5-9B on the WVS data. The table shows the accuracy, cross-entropy and KL divergence for the base model and various fine-tuned models on the training, validation and overall datasets. This is calculated by running the model with reasoning off against the modal response dataset.

3.2 Methods for creating the training dataset

We choose the WVS dataset (free, open and sufficient). Has other countries. NZAVS would be good yet no free access.

3.2.1 Clustering of survey data

The WVS data has 1,057 responses to 251 value questions. Each question has categorical responses with varying number of options. The goal of the clustering is to find groups of respondents that have similar responses to the value questions and sufficiently different responses to other groups of respondents. Furthermore the clusters should be robust to the choice of clustering method and reproducible.

To measure the effectiveness of the clustering we will use a few metrics including Bayesian Information Criterion (BIC) [17], cluster size entropy, within cluster coherence and inter cluster difference. Within cluster coherence is a measure of how dense the clusters response distributions are and is calculated as the entropy of the response distributions. Inter cluster difference is a measure of how different the clusters response distributions are from each other. These performance metrics are used to help understand how well the model has fit the data. We can use the Adjusted Rand Index (ARI) [7] to measure difference of cluster assignments. With this difference metric we can compare how stable a clustering is when retrained in various ways.

Clustering method

A standard method of clustering is to use Latent Cluster Analysis (LCA) [8, 1, 5]. This assumes that each respondent belongs to a latent class and that the responses are generated by a mixture of underlying latent classes. It has been used in social science research to cluster survey respondents [10]. Problematically it has several assumptions which are violated namely Local (conditional) Independence, Homogeneity and the number of classes has been correctly specified.

Given some of the standard assumptions of LCA are violated I compare various clustering methods in Table 3.2. We can see that LCA has the best inter cluster difference and within cluster coherence with sufficiently even cluster sizes for a 2 cluster situation. To further validate this in Table 3.3 we can see that LCA still has the best intra cluster coherence and near top performance in inter cluster differences in the 3 cluster situation. As a further basic check we can see in Figure 3.3 that the clusters of LCA and K-modes are similar but not identical.

Method	Cluster size entropy	Mean pairwise TVD	Within-cluster coherence
LCA	0.996000	0.194400	0.284000
K-Modes	0.999000	0.173500	0.279000
MCA+KMeans	0.181000	0.193200	0.265000
MCA+Ward	0.167000	0.191100	0.269000
MCA+KMedoids	0.172000	0.193500	0.268000
MCA+Spectral	0.167000	0.191100	0.269000
MCA+HDBSCAN	0.219000	0.147800	0.263000
MCA+SOM	nan	nan	nan
MCA+GMM	0.912000	0.088300	0.265000

Table 3.2: Comparison of different clustering methods with 2 clusters.

Given the above analysis we will use LCA as the clustering method for the WVS data. It is both a baseline in literature and performs well in preliminary analysis.

Method	Cluster size entropy	Mean pairwise TVD	Within-cluster coherence
LCA	0.987000	0.204100	0.295000
K-Modes	0.995000	0.186200	0.289000
MCA+KMeans	0.697000	0.212900	0.284000
MCA+Ward	0.190000	0.212700	0.265000
MCA+KMedoids	0.219000	0.210300	0.266000
MCA+Spectral	0.206000	0.198700	0.265000
MCA+HDBSCAN	0.219000	0.147800	0.263000
MCA+SOM	0.258000	0.131100	0.263000
MCA+GMM	0.888000	0.116200	0.271000

Table 3.3: Comparison of different clustering methods with 3 clusters.

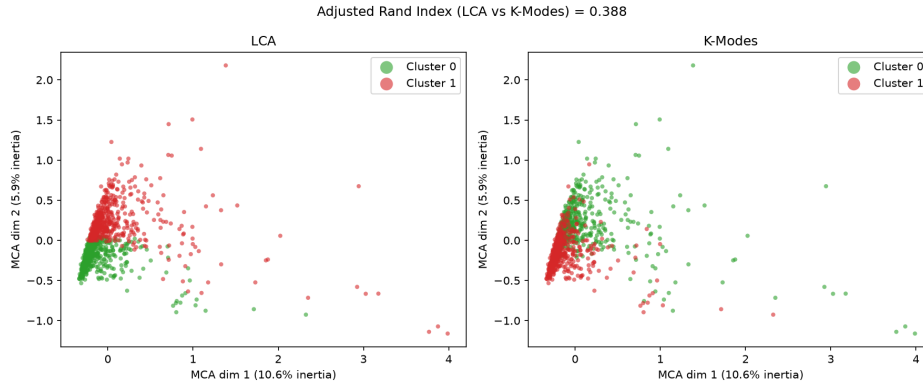


Figure 3.3: The clusters of LCA compared with the clusters of K-Modes. We can see that K-Modes and LCA produce different structured clusters.

LCA number of clusters selection

LCA clustering requires a predefined cluster size to a sweep was done to find the optimal number of clusters Figure 3.4. This indicates that 2 clusters is optimal. It is noted that BIC can unselect [13], therefore a Bootstrap Likelihood Ratio Test (BLRT) [11, 12] was used to test whether the additional cluster is significant. The test with $n = 30$ slightly rejected ($p = 0.032$) the null hypothesis that 2 clusters is sufficient in favour of 3 clusters. This is not a decisive result however begs further exploration.

The difference between $k = 2$ and $k = 3$ solutions is that the $k = 3$ solution splits cluster 1 of the $k = 2$ solution into two clusters which can be seen in Figure 3.5. The question is whether this split is meaningful or not. To answer this we can test the stability. This works by sampling data from the model and then refitting to that data and using ARI to compare the cluster assignments. As seen in Table 3.4 the $k = 2$ solution is much more stable than the $k = 3$ solution.

solution	mean ARI	95% CI
k=2	0.718	[0.572, 0.848]
k=3	0.467	[0.242, 0.704]

Table 3.4: Bootstrap stability check for 2 and 3 clusters. We can see that the $k = 2$ solution is much more stable than the $k = 3$ solution.

Given the instability of the $k = 3$ solution it is decided to use the $k = 2$ solution as it is more robust and reproducible.

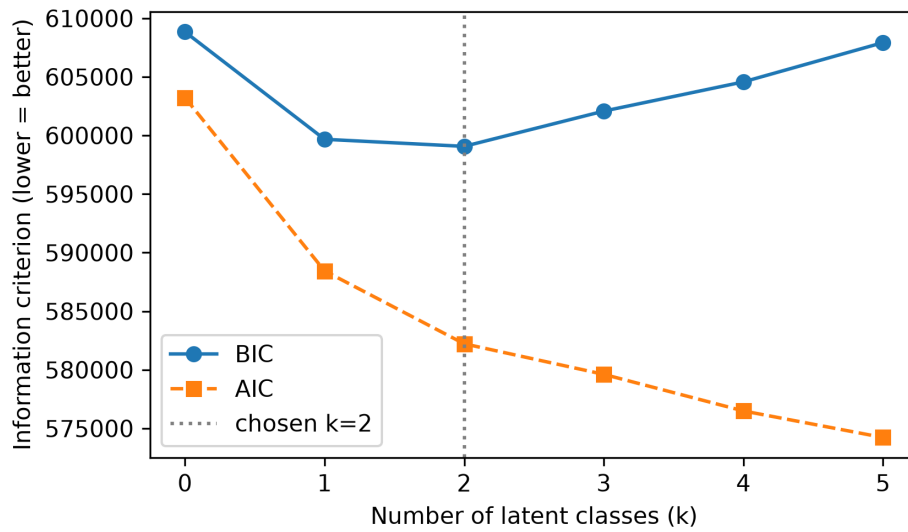


Figure 3.4: BIC values for different numbers of clusters. The optimal number of clusters (2) is indicated by the minimum BIC value.

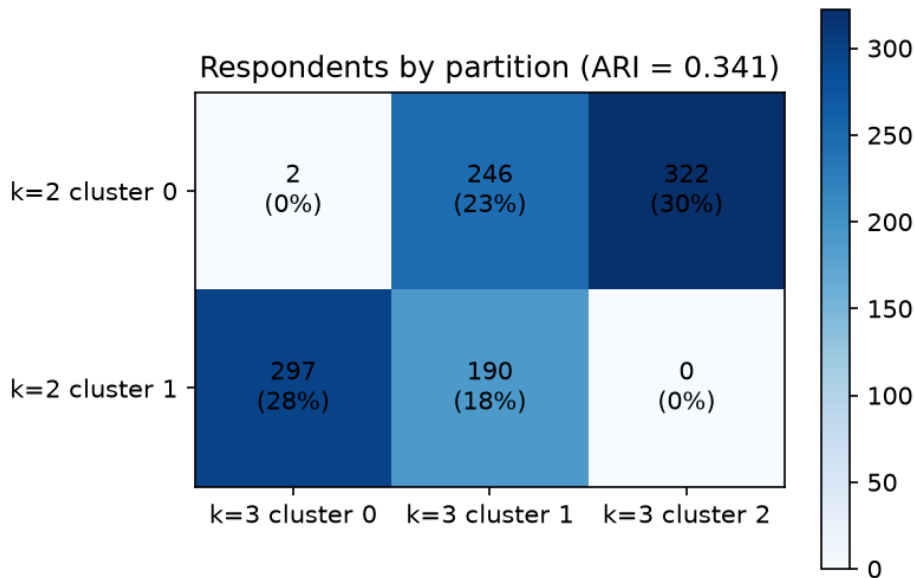


Figure 3.5: Cross-tabulation of the cluster assignments for 2 and 3 clusters. We can see that the $k = 3$ solution is simply splitting cluster 1 of the $k = 2$ solution into two clusters.

3.2.2 Structure of the training data

[3] introduce using WVS.

[9] mention how the order of options matter. We have to use WVS order to make it fair. Other research has showed it may not matter as we are fine tuning.

[16] show that LLM have the same human like biases. Therefore that is why we make it as similar as possible to original survey. Caveat is the matrix questions as only a single one is presented at each moment.

[15] show the forcing a model response format and paraphrases effect it a lot.

3.3 Methods for Fine-tuning the models

To better understand the effects on downstream behaviour 2 types of fine-tuning were done. First is modal fine-tuning where the goal is to have a model that responds in a way that is like the most common response in the survey data for that sub population. The second is distributional fine-tuning where the goal is to have a model that responds in a way that is like the distribution of responses in the survey data for that sub population.

Given the use of LLM reasoning in downstream applications allows for more uncertainty (citation needed, i.e the intuition that the reasoning gives it more chance of randomness and therefore can better model uncertainty).

There are ways of doing finetuning of a LLM. We will only explore simple fine-tuning methods (SFT) and vary the method on two axes: what the target is (modal response, sampled response or the distribution of responses) and what the response representation is (full option string or just single token representing the option).

The modal response fine tuning is conducted as a baseline as it is a straight forward application of SFT. The rest of the methods are various ways of trying to have the model learn the distribution of responses in the survey data. However in an implementation sense there are three methods (SFT, full option string distributional fine-tuning and single token distributional fine-tuning).

3.3.1 SFT fine-tuning: Modal & Sampled response

To do modal fine-tuning the model is trained used SFT with the expected text response being the full option string of a response. For modal fine-tuning the expected text response is the most common response in the survey data for that sub population. For sampled response fine-tuning the expected text response is a randomly sampled response from the survey data for that sub population.

[14] shows SFT used before RLHF

[18] shows that SFT is good enough.

3.3.2 LLM models

Definitions: V is the vocabulary size, The model parameters are θ . The model is a function f_θ that maps a prompt and a completion prefix to logits over the vocabulary.

Let a training example consist of a prompt (system + user) tokenised as $x = (x_1, \dots, x_p)$ and a completion (assistant answer) tokenised as $y = (y_1, \dots, y_T)$, with T the completion length.

The model is autoregressive, so the probability of the completion factorises over tokens:

$$p_{\theta}(y \mid x) = \prod_{t=1}^T p_{\theta}(y_t \mid x, y_{<t}), \quad y_{<t} = (y_1, \dots, y_{t-1}). \quad (3.1)$$

The probability of the next token is given by a softmax over the logits:

$$p_{\theta}(y_t \mid x, y_{<t}) = \frac{\exp f_{\theta}(x, y_{<t})_{y_t}}{\sum_{v=1}^V \exp f_{\theta}(x, y_{<t})_v}. \quad (3.2)$$

3.3.3 Supervised fine tuning loss function

SFT fits the model by *maximum likelihood* on the demonstration data: maximising $\log p_{\theta}(y \mid x)$ is equivalent to minimising the negative log-likelihood (NLL),

$$\mathcal{L}_{\text{SFT}}(\theta; x, y) = -\log p_{\theta}(y \mid x) = -\sum_{t=1}^T \log p_{\theta}(y_t \mid x, y_{<t}), \quad (3.3)$$

and the objective over the dataset $\mathcal{D}$ is $\theta^* = \arg \min_{\theta} \sum_{(x,y) \in \mathcal{D}} \mathcal{L}_{\text{SFT}}(\theta; x, y)$.

Each per-token term in (3.3) is a categorical cross-entropy: with $z_t = f_{\theta}(x, y_{<t}) \in \mathbb{R}^V$ the logits over the vocabulary of size V ,

$$-\log p_{\theta}(y_t \mid x, y_{<t}) = -z_{t,y_t} + \log \sum_{v=1}^V \exp(z_{t,v}). \quad (3.4)$$

3.3.4 Distributional loss functions

Rather than modelling a single completion y instead we can model distributions over the K possible completions of a prompt¹. Therefore the objective is to minimise the difference between the models predicted distribution p and the expected distribution q .

$$\mathcal{L}_{\text{distributional}}(\theta; x, y) = -\sum_{i=1}^K q_i * \log \tilde{p}_{\theta}(y_i \mid x) \quad (3.5)$$

where $\tilde{p}_i$ is the softmax over the K completions of the options log-probs.

$$\tilde{p}_{\theta}(y_i \mid x) = \frac{\exp(\log p_{\theta}(y_i \mid x))}{\sum_{j=1}^K \exp(\log p_{\theta}(y_j \mid x))} \quad (3.6)$$

There are two ways in which we use this loss function. The first is that y is a full string of the response and the second is that y is a single token representing the response.

We can see that the distributional loss function is a generalisation of the SFT loss function. If we have a single expected response that is one-hot encoded the distributional loss function reduces to the SFT loss function (3.3) with an constant offset of $\log \sum_{j=1}^K \exp(\log p_{\theta}(y_j \mid x))$.

Full option string distributional fine-tuning

When $T > 1$ to get the $p_{\theta}(y_i \mid x)$ for each of the K completion each example per batch is expanded to K examples with the same prompt and each of the K completions. The loss function is then calculated as in (3.5).

¹This is possible as the task is a classification task with a finite number of possible completions.

Single token distributional fine-tuning

Rather than using the full option string we can instead augment the training data to make the expected response a single token (e.g. A, B, C, D, E) representing the response. The model is then trained to predict the distribution of responses over the single token representation of the responses.

This is the simpler of the two and used in other research [4, 2]...

Chapter 4

Simulation

A section discussion the behavioural simulations of the model

The selection of scenarios The harness used How summaries of the results were generated
A method for mapping to value dimensions The results of the simulations (compared to clusters are there behavioural clusters)

Chapter 5

Survey

5.1 Results of the survey

The public did prefer models that were aligned with their responses... Or maybe they don't. We are still to find out.

5.2 How the survey was constructed

5.2.1 Survey design

5.2.2 Assignment of respondents to clusters

At the start of the survey a handful of questions will be asked to determine which cluster the respondent belongs to. The questions selected are most information questions to help assign a respondent to a cluster.

Some questions are considered controversial and so were omitted from potential cluster assignment questions. As seen in Figure 5.1, this has little effect on the confidence of the assignment. In the final 10 questions selected

Add appendix showing questions selected.

we can assign cluster with a mean of 0.915 and over 74% of the WVS respondents with over 90%.

5.2.3 Selection of scenarios

5.3 Analysis of the survey results

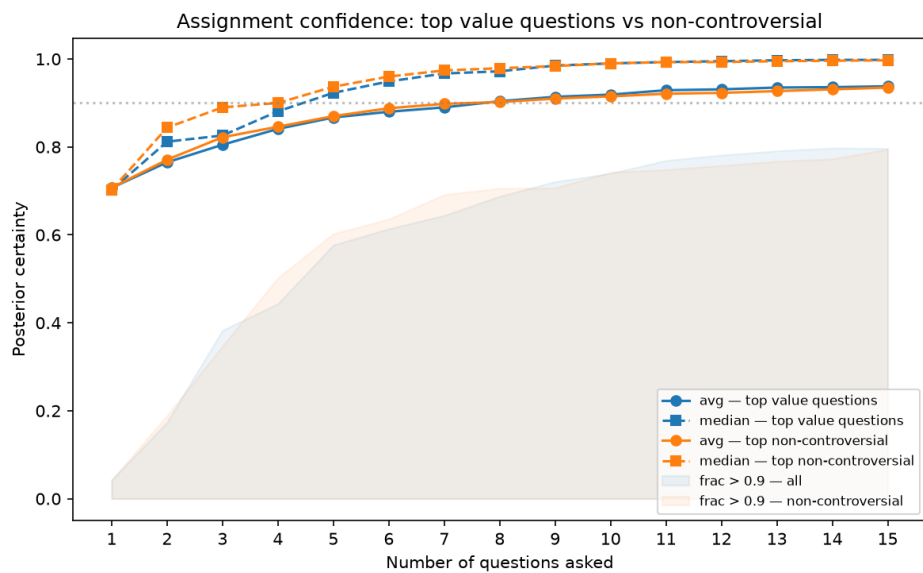


Figure 5.1: The confidence that we can assign a respondent accurately to a cluster by how many of the most informative questions they answer.

Chapter 6

Conclusions

The conclusions are presented in this Chapter.

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Appendix A

Project details

A.1 Budget and Resources

Funding (received)	
Blue dot impact rapid grant	2542
VUW programme	500
VUW AI centre	1000
Total funding	4042
Expenses	
GPU compute (160 hours at \$2.00/hour)	320
Participant compensation (500 participants at \$5.25 each)	2625
Participant recruitment admin fee	500
Contingency (20% of direct costs)	689
Total expenses	4134
Balance (funding – expenses)	-92

Table A.1: Detailed project budget in New Zealand Dollars (NZD).